

EcoTrack

Satellite-based estimation of urban fossil-fuel CO₂ emissions
Shivajinagar → Talegaon Dabhade corridor, Pune

Progress report: what has been done, what was found, and what comes next

Janhavi Tupe · Final-year project · October 2026

Headline so far: Pune + Pimpri-Chinchwad emit about **2.79 million tonnes of fossil CO₂ per year** (annual mean; 95% range 1.79–4.36), estimated from satellite NO₂, CO and wind data. One of the two official inventories (EDGAR, 3.6) agrees; the other (ODIAC, 11.8) is ruled out.

1. The problem and the idea

Cities produce most fossil CO₂, but the official numbers (emission *inventories*) are slow, coarse and often disagree. For Pune, two respected inventories differ by a factor of 3.2. Measuring CO₂ directly from space is hard: Pune adds only ~0.1 ppm to a ~420 ppm background, far below the noise of today's CO₂ satellites.

The idea (from the base paper, Xie et al. 2026): burning fuel releases CO₂ *and* NO₂. NO₂ is easy to see from space (low background, lives only ~1–2 hours, measured daily by TROPOMI). So measure the city's NO₂ emission, then convert it to CO₂ with a CO₂:NOx ratio. That ratio is the weak link, and the base paper's largest error term.

Research question: can combining several satellites (NO₂ + CO + CO₂), weather and human-activity data estimate urban CO₂ better than relying on NO₂ alone?

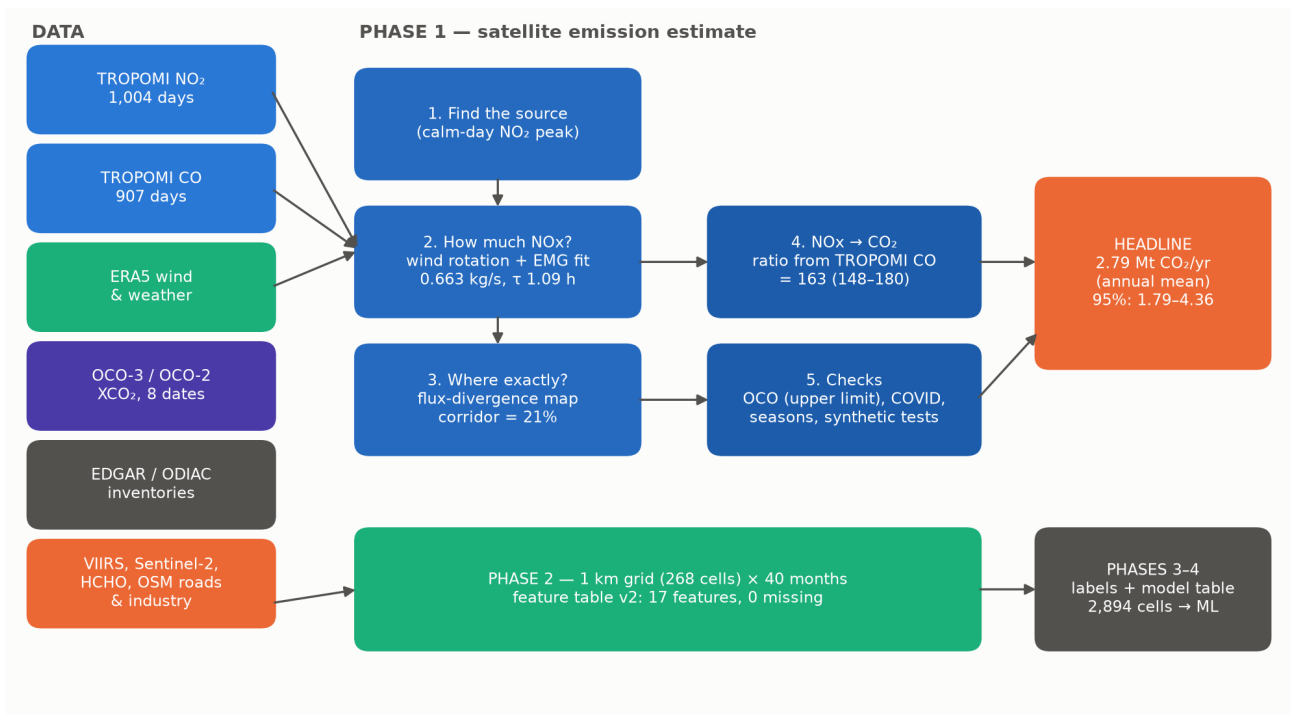


Figure 1. The whole workflow so far: data → Phase 1 emission estimate → checks → headline; Phase 2 builds the 1 km feature table for the machine-learning phases.

2. Study area and data

A 33 km strip along the Old Mumbai–Pune Highway where factories (MIDC Bhosari, Chinchwad, Talegaon) and heavy traffic sit side by side. The corridor is the highway centreline ± 3.5 km plus Talegaon MIDC (269 km²), divided into 268 grid cells of 1 km and three clusters.

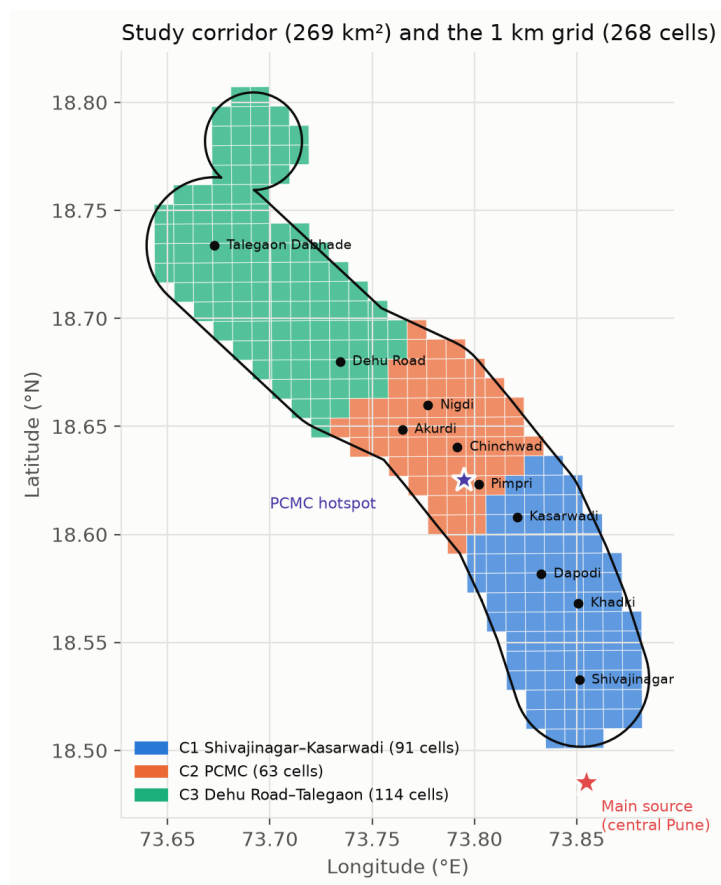


Figure 2. The corridor, its 1 km grid coloured by cluster, the ten waypoints, the main emission source found in Phase 1 (central Pune, just south of the corridor) and the second hotspot (Pimpri–Chinchwad).

Dataset	What it measures	Used for
TROPOMI (Sentinel-5P)	NO ₂ , CO, HCHO columns; daily, $\sim 3.5 \times 5.5$ km	Emissions, sector mix, chemistry proxy
OCO-3 / OCO-2 (NASA)	Column CO ₂ (XCO ₂); 8 usable dates over Pune	Direct CO ₂ check
ERA5 (ECMWF)	Hourly wind, boundary layer, temperature, sunlight	Moving the plumes; features
EDGAR, ODIAC	Bottom-up emission inventories	CO ₂ :NO _x prior, comparison (never truth)
VIIRS, Sentinel-2	Night lights; NDVI/NDBI	Human-activity features
OpenStreetMap / GRIP4	Roads, places, industrial land	Corridor geometry, road density
SRTM	Ground height	Terrain correction for CO

Study period: October 2019 – May 2024, October–May only (five dry seasons, 40 months; D7). It starts after TROPOMI's pixel-size change (August 2019) and avoids monsoon cloud. There is no monsoon season, so no full annual cycle is observed.

Quality control: TROPOMI pixels with cloud fraction above 0.3 or a sun more than 70° from overhead are dropped; an overpass is used only if at least half the corridor is valid. Remaining outliers are removed per pixel against its own 5-year history (Tukey far-out fences, $k = 3$; D13). This removes 0.09% of pixels and

changes every result by less than 2%. Result: 1,004 usable NO₂ days and 907 CO days.

3. First: checking the data exist (feasibility, G1–G4)

Before building anything, four go/no-go checks with pass rules set in advance:

Check	Result	Verdict
G2 TROPOMI coverage	Median 26 usable days per month; worst month 13	PASS
G3 NO ₂ signal	All clusters 1.7–2.4× above clean background, every season	PASS
G1 OCO-3 / OCO-2	Pune is an OCO-3 'snapshot' target; 8 strong dates	Case study only
G4 Waypoints	6 of 10 were > 500 m off (Dehu Road 4.4 km); corridor rebuilt on the real highway	FIXED

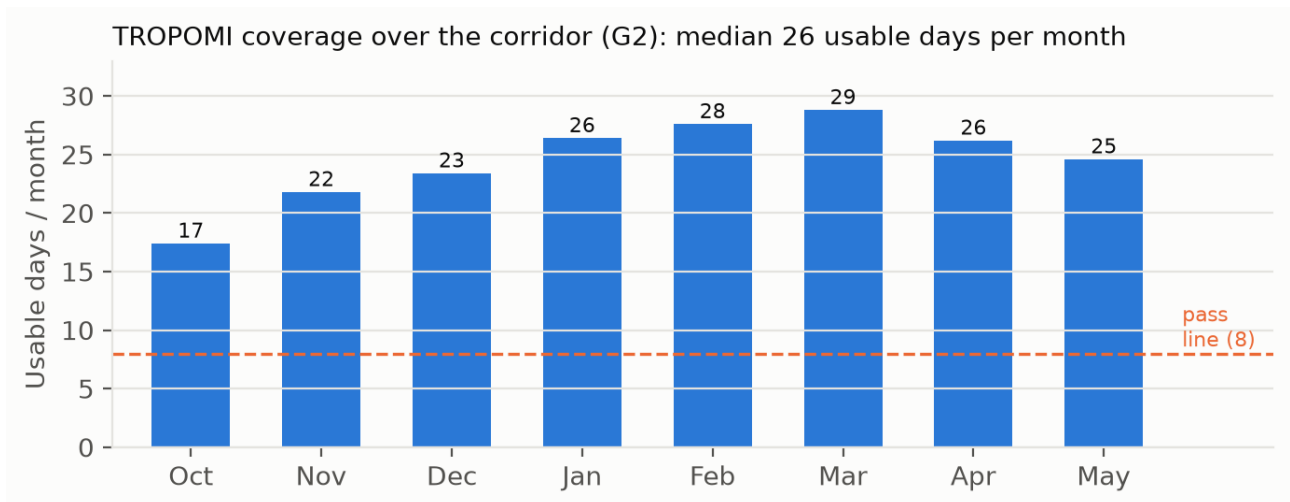


Figure 3. Usable TROPOMI overpasses per month (5-season average). October is weakest (late-monsoon cloud); every month clears the pass line comfortably.

4. Phase 1: measuring the city's emissions from space

Step 1, find the source. On calm days NO_2 piles up over its source. The calm-day average peaks over central Pune (Swargate/Camp), just south of the corridor.

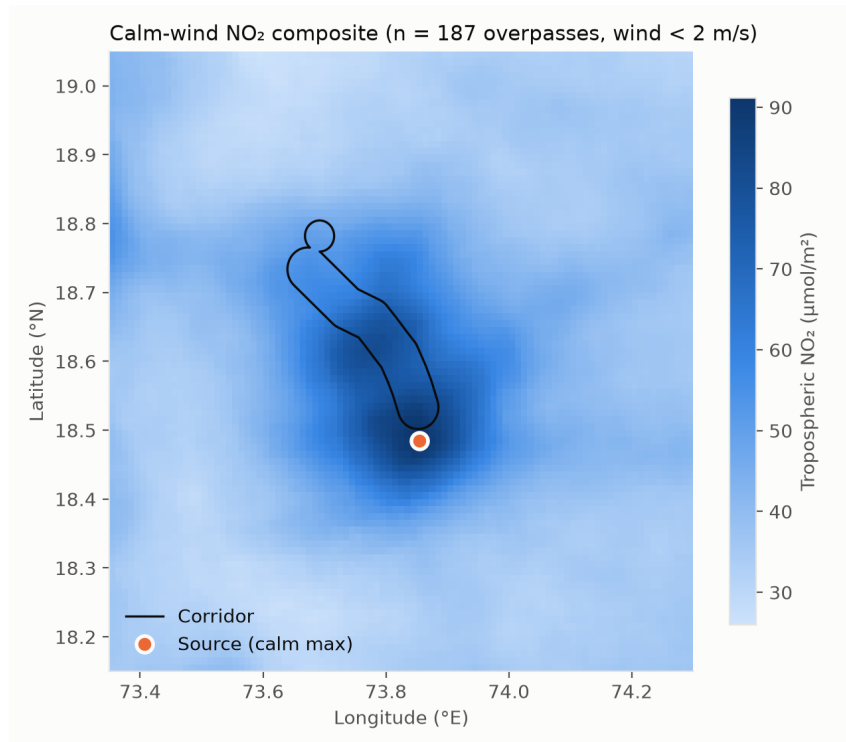


Figure 4. Average NO_2 on 187 calm days. Orange dot: the main source.

Step 2, how much? Each day's image is rotated so its wind points the same way, then averaged; summing across the wind gives the *line density*. A plume curve (EMG) is fitted, giving the NO_2 in the plume and how far it travels before decaying. Result: **0.663 kg/s NO_x** [95% CI 0.620–0.715], lifetime **1.09 h**.

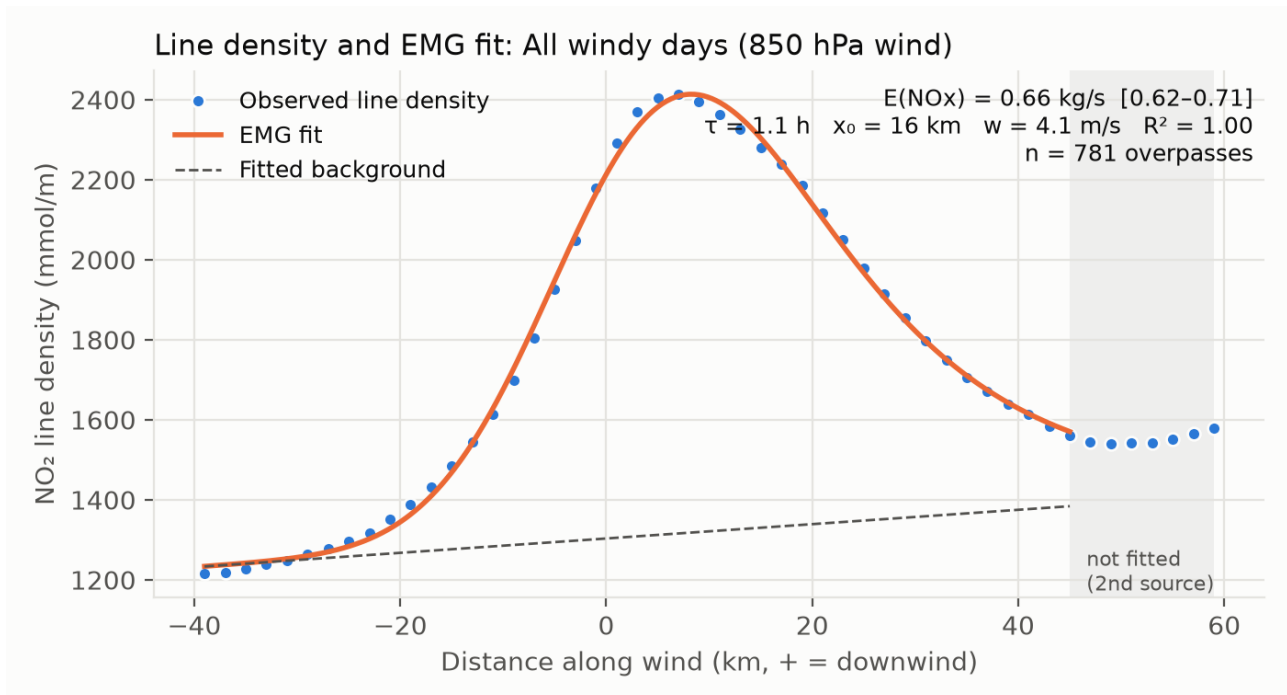


Figure 5. Line density (blue dots) and the fitted plume curve (orange), fitted up to 45 km downwind. The grey band is excluded because a second source sits there (see section 6).

Details of the fit. Winds are ERA5 at 850 hPa (about 1.5 km up, inside the mixed layer), matched to each overpass hour. Days with 2–8 m/s wind are used (781 overpasses); calm days (< 2 m/s) are kept for Step 1. The rotated images are binned at 2 km, summed ± 20 km across the wind, and fitted from 40 km upwind to 45 km downwind. NO₂ is converted to NO_x with the standard factor 1.32. The fit uses Differential Evolution (a global optimiser: 300 candidate solutions, mutation 0.5–1.0, crossover 0.7) and 200 bootstrap resamples of the days for the 95% interval. A test of 36 optimiser settings found that every converged run lands on the identical answer (0.6627 kg/s), so the result does not depend on the settings.

How robust is it? The same fit on subsets of days and with other choices:

Run	Days	NOx (kg/s) [95% CI]	Lifetime (h)
Main: 850 hPa wind, 2–8 m/s, $\leq$ 45 km, sloped background	781	0.663 [0.620–0.715]	1.09
Easterly winds only / westerly winds only	429 / 276	0.600 / 0.484	1.53 / 1.94
Light (2–4 m/s) / stronger (4–8 m/s) wind	416 / 365	0.567 / 0.695	1.47 / 0.84
Wind taken at 100 m / at 10 m	782 / 675	0.673 / 0.556	0.95 / 1.05
Flat background 45 km / sloped 30 km / original flat 60 km	781	0.591 / 0.651 / 0.522	1.46 / 0.85 / 1.67

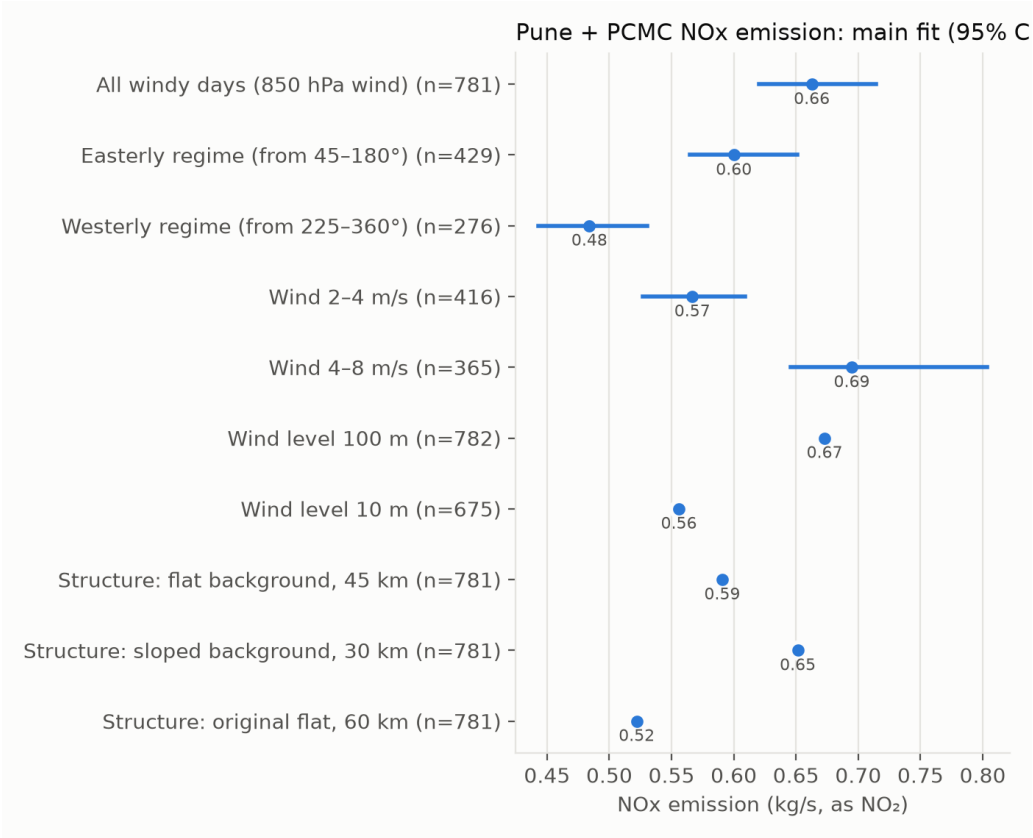


Figure 6. Every variant of the fit with its 95% interval. The spread between easterly and westerly days and the background/window choice are carried into the uncertainty budget.

Step 3, where exactly? A second method (flux divergence) maps emission pixel by pixel: what flows out minus what flows in, plus what chemistry destroys. It finds two hotspots 16.7 km apart: central Pune and Pimpri–Chinchwad. **The corridor produces about 21% of the metro's NOx**, a share stable to $\pm 2.4\%$ in every test.

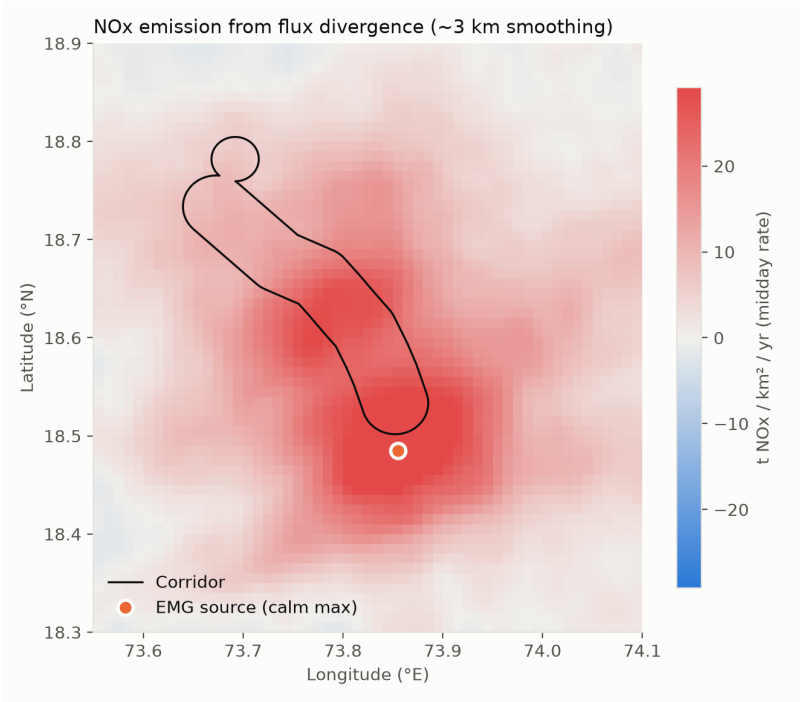


Figure 7. Emission map from flux divergence. Red = emitting.

Zone	Share of NOx within 25 km
Central Pune core	27.6%
Whole corridor	21.0% (stable to $\pm 2.4\%$ in every test and season)
C1 Shivajinagar–Kasarwadi	9.8%
C2 PCMC	6.3%
C3 Dehu Road–Talegaon	4.9%

Do the two methods agree? Flux divergence totals grow with the radius drawn around the source; at 25 km they give 0.787 kg/s, 19% above the plume fit. Totals depend on the assumed lifetime, so flux divergence is used for *where* (shares), and the plume fit for *how much* (D10).

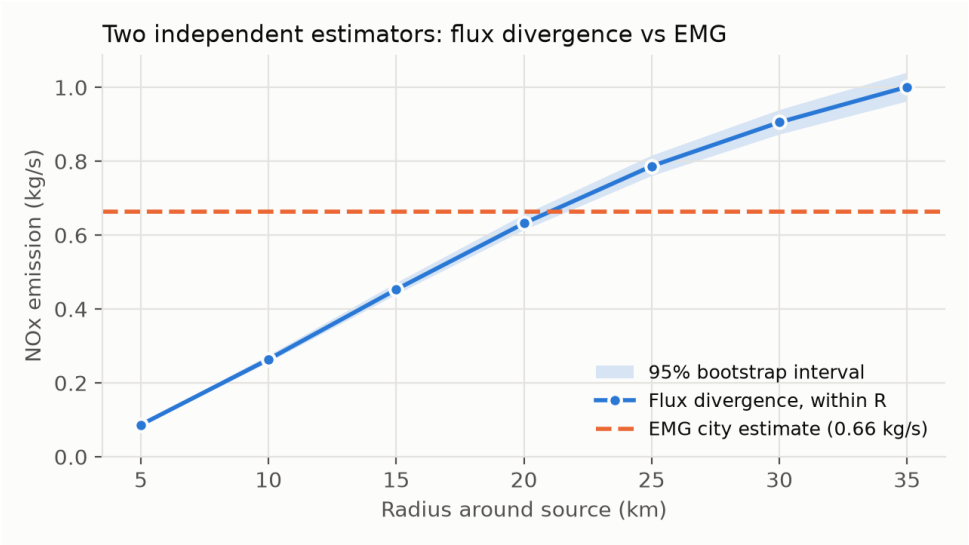


Figure 8. Flux-divergence total vs radius (blue) against the plume-fit city total (orange).

5. From NOx to CO₂, and the inventories

NOx is turned into CO₂ with a CO₂:NOx ratio. Inventories say it varies by sector from 115 (industry) to 393 (households), so the city's mix matters. **TROPOMI CO measures that mix:** CO does not decay, so downwind of the city it steps up and stays up. After removing the Western Ghats terrain imprint, the step gives a CO:NOx of **16.7**, close to the inventory's 16.0, and narrows the CO₂:NOx ratio to **163 (148–180)**.

TROPOMI CO: downwind step over Pune (terrain-normalised, static pattern removed)

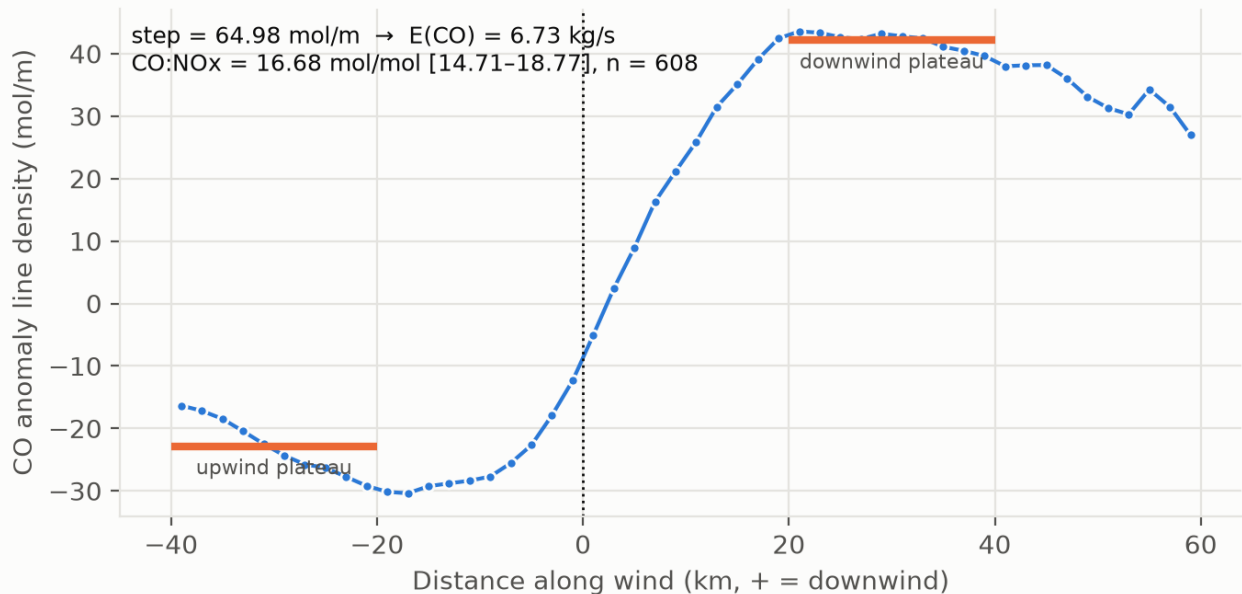


Figure 9. The CO 'step' over Pune: upwind plateau (left) to downwind plateau (right).

This is the core answer to the research question so far. With the inventory ratio alone (the base paper's approach), the ratio is a 25% error term. Measuring the sector mix with TROPOMI CO cuts it to about 14% (scenario spread 9.9% plus an assumed 10% for the inventory's per-sector ratios), and the total from 30.9% to 23.0%.

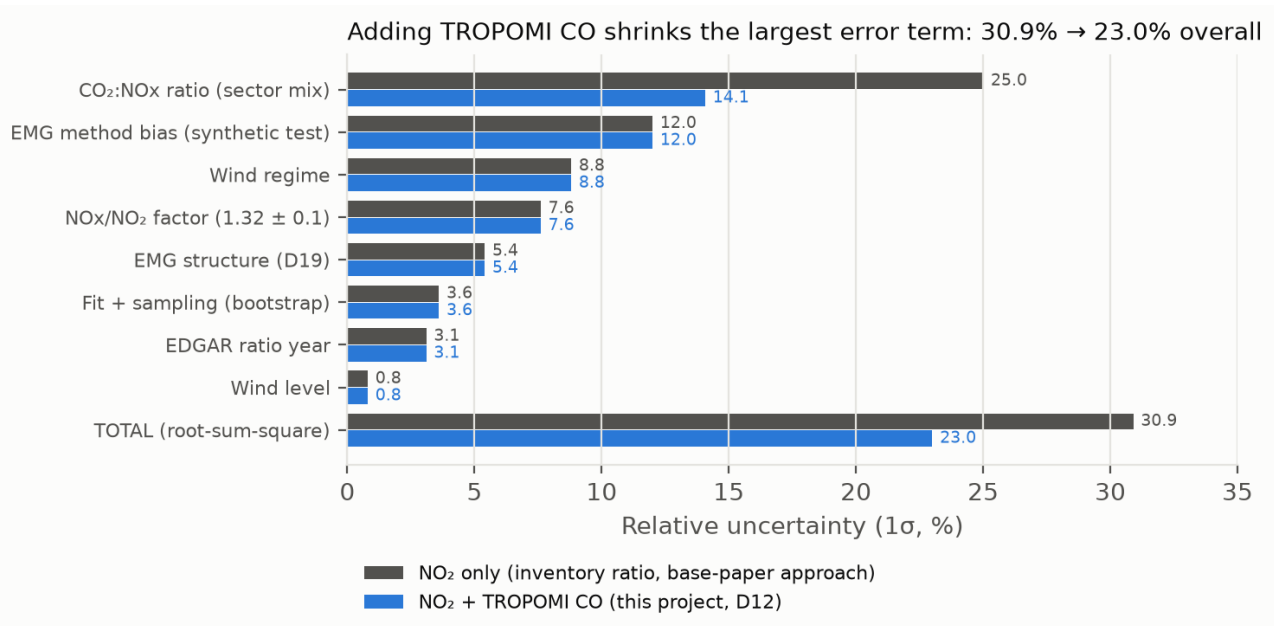


Figure 10. Uncertainty budget term by term, before and after adding TROPOMI CO.

From midday to a yearly figure. TROPOMI sees Pune at about 13:30 on October–May days, when traffic and industry are busy. Published hour-by-hour, weekday and monthly emission profiles (Crippa et al. 2020), weighted by the actual overpass days, give a factor $F = 1.20\text{--}1.23$. Dividing by it gives the annual mean: NO_x 17.0–17.5 kt/yr (EDGAR 21.1).

Uncertainty was propagated with a Monte Carlo (200,000 random draws of every input):

Fossil CO ₂ (Mt/yr)	Median	68% range	95% range
Midday October–May rate	3.39	2.76–4.16	2.26–5.08
Annual mean (headline)	2.79	2.23–3.50	1.79–4.36
Annual mean, with the measured +12% method bias removed	2.51	2.05–3.05	1.69–3.69

Monte Carlo uncertainty (200,000 draws): median, 68% (thick) and 95% (thin) ranges

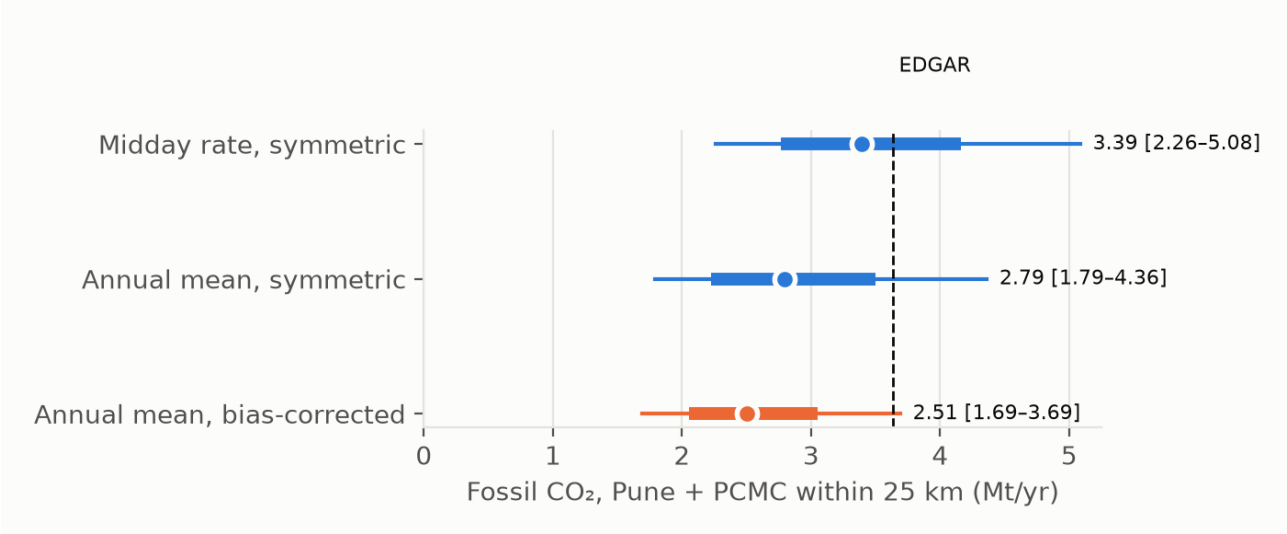


Figure 11. Fossil CO₂ for Pune + PCMC: median, 68% and 95% ranges. EDGAR (dashed) is inside; ODIAC (11.8, off the chart) is far outside.

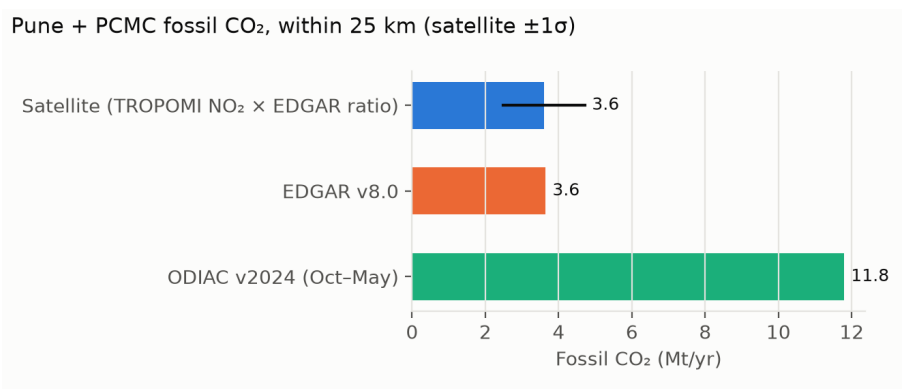


Figure 12. The two inventories disagree 3.2× for the same area (ODIAC units were checked: tonnes of carbon, converted ×44/12). The satellite estimate sides clearly with EDGAR.

Can CO₂ satellites confirm it? Not yet: Pune's CO₂ plume (~0.02–0.15 ppm) is 10–20× smaller than the noise and scan-stripe artefacts of OCO-3/OCO-2. The 8 snapshots are consistent with our estimate but only set an upper limit (about 7–14 Mt/yr). A scaling factor β (observed ÷ predicted plume) comes out 0.81 ± 0.74 (EDGAR implies 1.01): consistent, but too uncertain to be a measurement (D11). **So multi-satellite data help through CO, not through direct CO₂.**

OCO soundings (colour) and predicted Pune plume (contours 0.05 / 0.15 / 0.3 ppm)

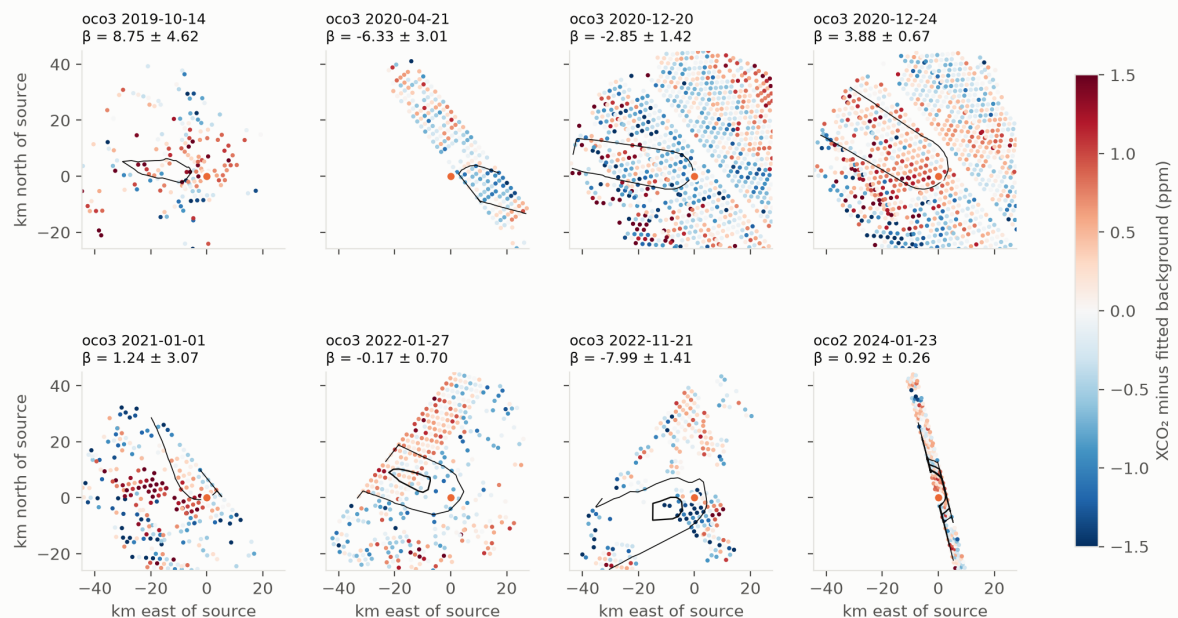


Figure 13. OCO soundings on the 8 dates (colour) vs the predicted Pune plume (contours). The stripes are instrument artefacts larger than the plume.

A second, independent emission map from CO. The flux-divergence method also works on CO once the terrain imprint is removed: its 25 km total is within 3% of the CO step, and the map matches the NO₂ map ($r = 0.75$). The corridor holds **14.5% of the city's CO but 21.0% of its NO_x**, an early sign of an industrial corridor (more NO_x per CO) versus a residential core. This map does not use NO₂ at all, which matters for Phase 3.

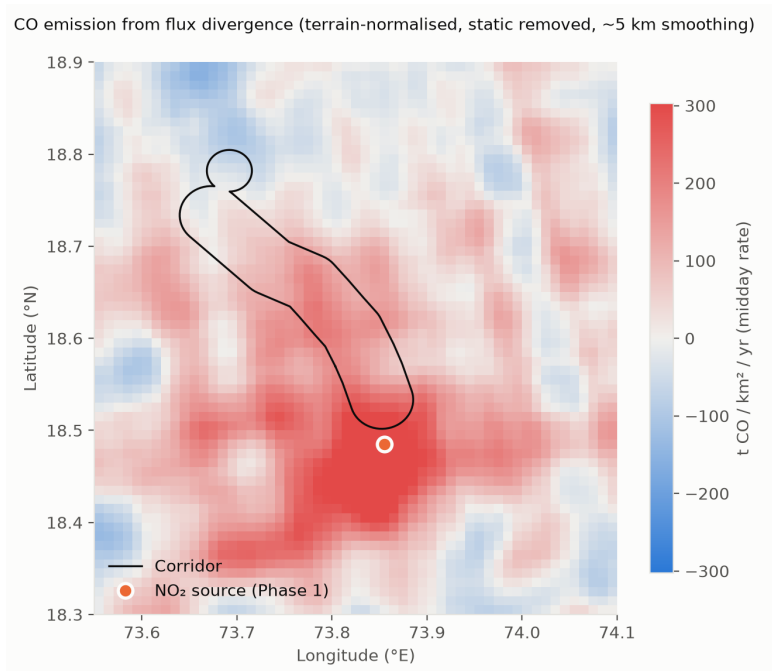


Figure 14. CO emission map from flux divergence (terrain-normalised).

6. Testing the method, and a correction it caught

Every method was first run on **fake data with a known answer**: a simulated city on the real grid with real-like winds and noise (13 automated tests in tests/):

Method	Recovered vs truth	Notes
Plume fit (EMG)	Emission +11 to +12%; lifetime -16%	Same in both random seeds, so systematic. Causes: across-wind window ~2%, mixing wind speeds ~6%, 2 km bins ~4%. Kept in the uncertainty budget.
Flux divergence	-4%	Stable at 15, 25 and 35 km; no false sources
CO step	-1.4%	Terrain alone gives ≈ 0 after correction
Outlier filter (D13)	Line density changed 0.9%	Catches 92% of injected spikes
CO flux divergence	Passes with Pune's real winds	With random winds it passed too easily; the real-wind test exposed a -70 mol/s terrain artefact, now removed

A robustness test (letting the background slope) then changed the answer by 42%. Investigating showed the original fit reached 60 km downwind, where **PCMC and Talegaon add a second source** that the single-source model misread. Fitting only to 45 km fixed it: both background shapes agree, and NO_x rose ~20%. Everything downstream was rerun, and one earlier conclusion that depended on the low value ("more household burning than the inventory") was withdrawn. The fix had its own side effect: on small subsets the 6-parameter fit became unstable, so a quality rule sends such fits (lifetime < 0.6 h or a bootstrap range wider than $\times 2$) back to a flat background.

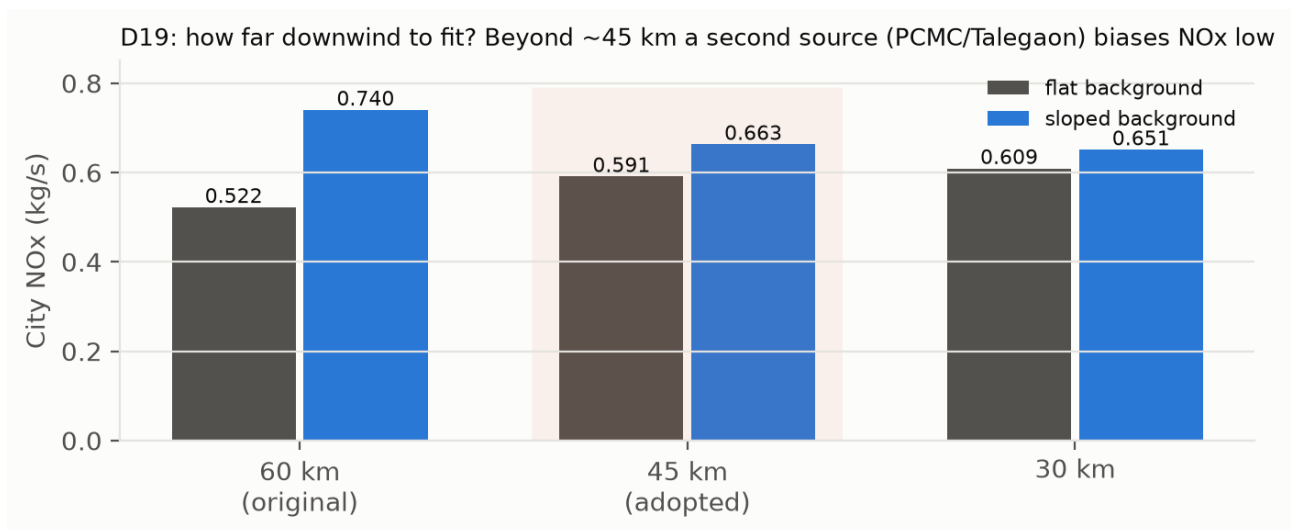


Figure 15. City NO_x by fit window and background shape. Inside 30–45 km the two agree (decision D19).

Does the method see real changes? A model-free check (city minus rural NO₂) for late March–May of each year:

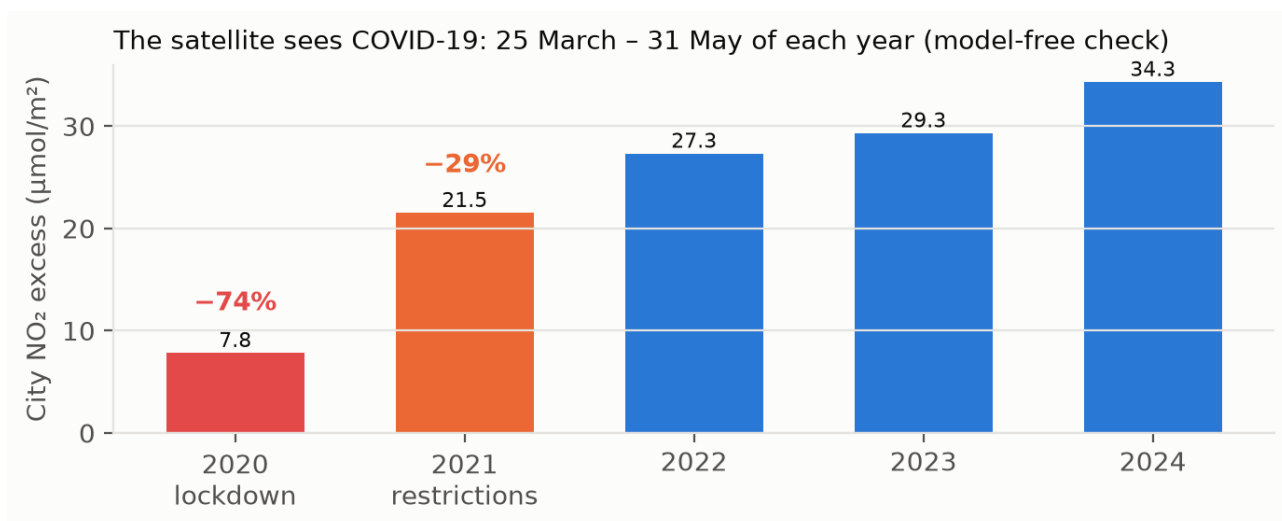


Figure 16. The 2020 national lockdown (-74%) and the 2021 Maharashtra restrictions (-29%) are both detected, in the right order, plus growth to 2024.

(The plume fit on the 2020 lockdown weeks alone was rejected: R^2 0.62, a parameter at its limit and an 8.8 h lifetime. When a model's assumptions fail, a simpler model-free measurement is used instead.)

Seasons. NOx can be resolved season by season (differences between seasons are 2.8× the uncertainty within one): 0.43 kg/s in 2019–20, rising to 0.60–0.73 afterwards. The corridor's share stays between 20% and 22.4% in every season. Winds reverse with the season: east-southeast in October–March (Pune's plume blows up the corridor), west-northwest in April–May.

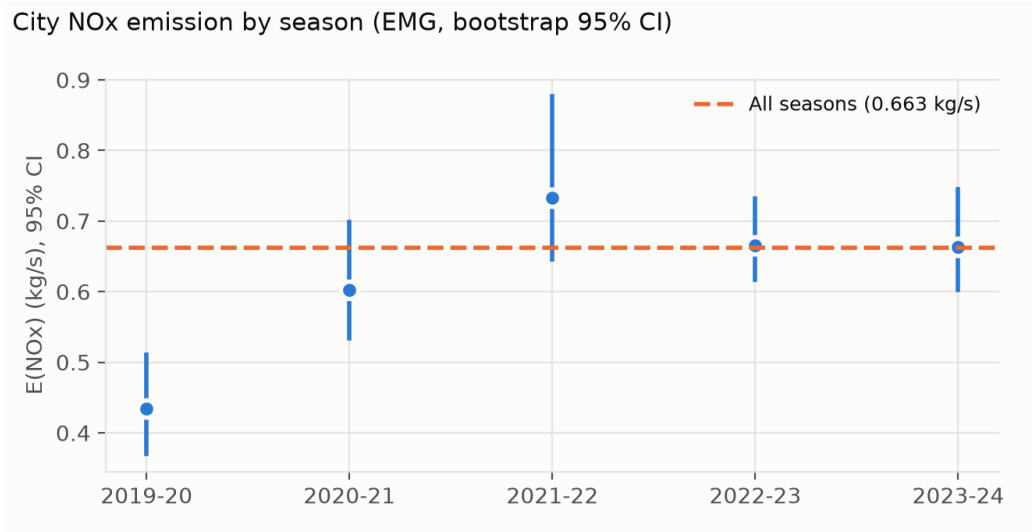


Figure 17. City NOx per season with 95% intervals; dashed: all seasons together.

7. Phase 2: the 1 km feature table

For the machine-learning phases, every layer was put on the 1 km grid, one row per cell per month: 268 cells × 40 months = **10,720 rows, 17 features, 0 missing values** (feature table v2, frozen with a SHA-256 fingerprint). The features are NO₂, CO, HCHO (a chemistry proxy), wind and weather, night lights, vegetation and built-up indices, road density and industrial-land fraction. Inventories are kept only as reference columns, never as inputs.

Four of the 17 features on the 1 km grid (average over 40 months)

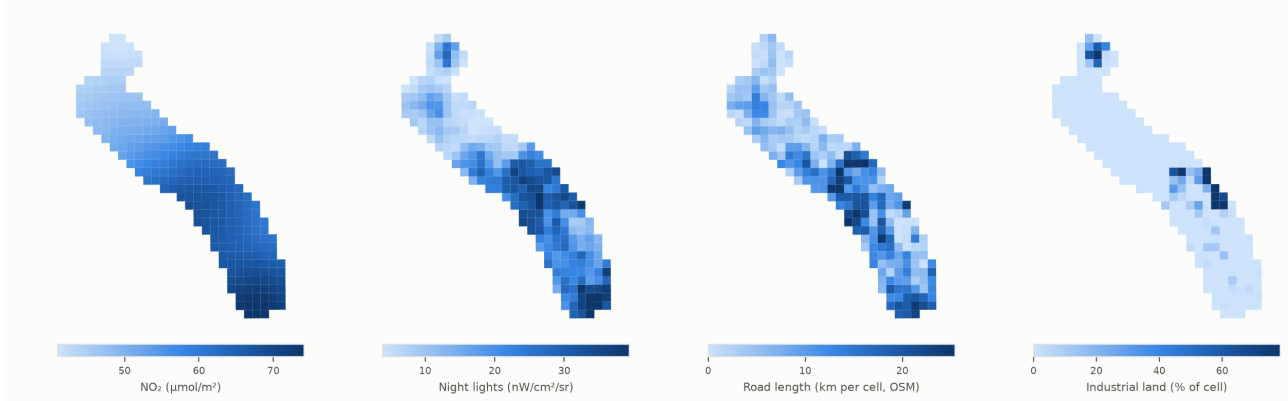


Figure 18. Four features averaged over 40 months. NO₂ is highest near Pune; night lights and roads trace the urban core and PCMC; industrial land picks out MIDC Bhosari, Chinchwad and Talegaon MIDC.

Roads and industry from OpenStreetMap. The OpenStreetMap query servers were overloaded for days, so the whole western-India map file (221 MB, checksum-verified) was downloaded once and read locally in 2.5 minutes. It gives 36,082 roads and 223 industrial areas. A second published road map (GRIP4) agrees on major roads (rank correlation 0.84) but misses almost all local streets (0.07 vs 4.2 km per cell), so OpenStreetMap is the better source.

A check before any machine learning: which inputs can say where? For each feature we measured how much of its variation is between cells rather than between months. Roads, industry, land cover and night lights vary in space; CO, HCHO and weather barely differ between 1 km cells. So at this scale the location of emissions must come from the activity layers (and partly NO₂), while CO and weather help with timing. This is written down *before* training, and it is why the models will be tested on held-out map areas (spatial cross-validation).

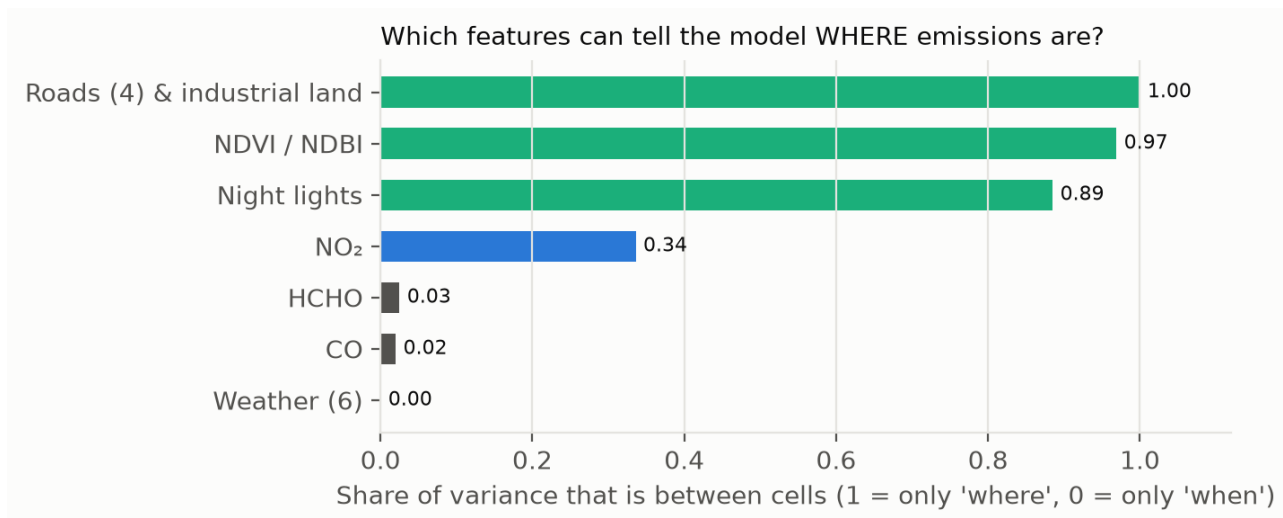


Figure 19. Share of each feature's variance that is spatial (between cells).

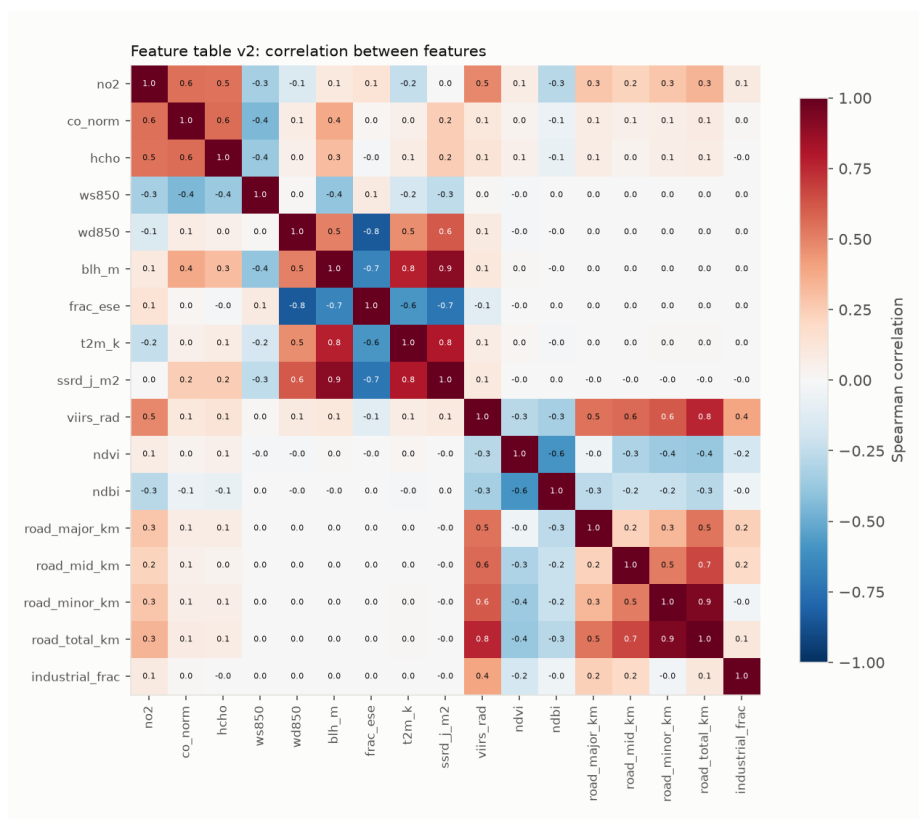


Figure 20. Correlation between features. Only expected pairs are strongly related (minor vs total roads; boundary layer, sunlight and temperature; two wind descriptors). Industrial land is nearly independent of everything.

A caveat found in QA: the 'built-up index' NDBI is highest in rural Talegaon and is negatively related to NO₂ and night lights. In the dry season, bare soil reflects infrared like concrete, so NDBI is not a reliable built-up measure here; roads, industrial land and night lights are the dependable activity layers.

8. Phase 3: the CO₂ labels

The machine-learning phases need a CO₂ value (a *label*) for every cell and season. There is no ground truth at 1 km, so labels must be built, and the main risk is **circularity**: if a label is built from a layer that is also a model input, the model just learns the recipe back and looks better than it is (D2). The proposal's activity-weighted label had exactly this problem for Experiment D.

What was built (labels v1, design D16, awaiting the guide): 268 cells × 5 seasons = 1,340 labels, two per row, each with an uncertainty from bootstrapping the satellite days:

	L-fd (primary)	L-co (independent)
Built from	NO ₂ emission map × seasonal city NO _x × 163	CO emission map × city CO ₂ (no NO ₂ in the pattern)
Varies by	cell and season	cell only (D21)
Median (t CO ₂ /km ² /yr, midday)	2,374	2,010
Share of the city in the corridor	20.7%	14.3%
Noise ceiling (best possible R ²)	0.97	0.75
Correlation with the NO ₂ feature	0.91 (its recipe)	0.79 (never used NO ₂)

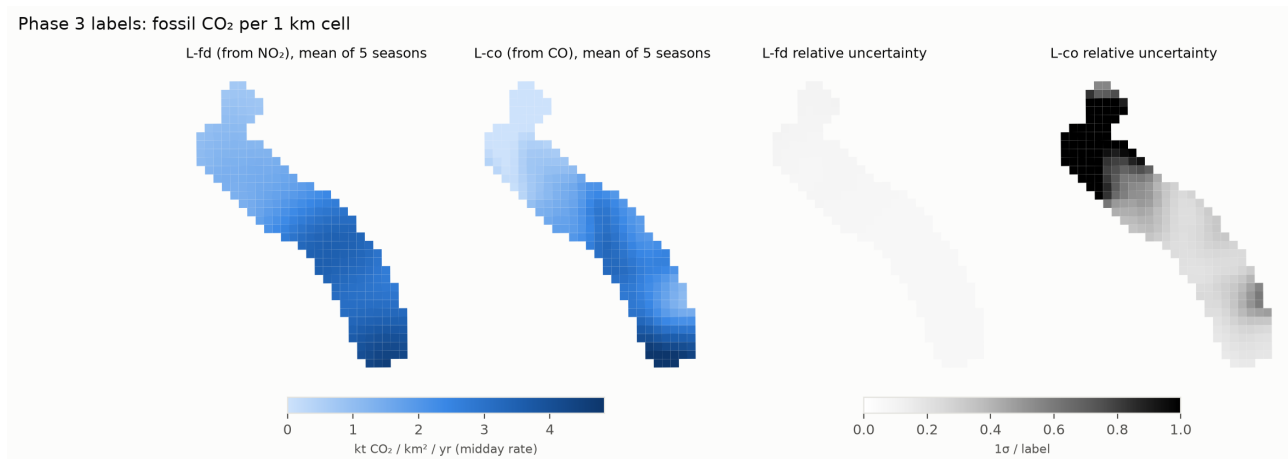


Figure 21. The two labels (multi-year mean) and their relative uncertainty. Both show the gradient from Pune up the corridor; L-co is weak and noisy in rural Talegaon.

Checks: the rebuilt NO₂ map equals the Phase 1 map exactly; the cells reproduce the corridor shares (20.7% vs 21.0%; 14.3% vs 14.5%); changing the smoothing scale barely matters ($r \geq 0.98$); each season's NO₂ map matches the 5-year pattern ($r \geq 0.98$), so one fixed pattern is justified. The two labels agree on *where* (cell $r = 0.93$).

A decision forced by the data (D21): CO's seasonal totals vary less than their own noise (between/within 0.49, against 2.8 for NO₂, the same rule as Phase 1). So the CO label is spatial-only: it tests whether a model finds *where* emissions are; the NO₂ label tests *where and when*.

Circularity, measured: the NO₂-built label correlates 0.91 with the NO₂ feature, so with it Experiment A is a 'construction baseline'. But the CO-built label, which never used NO₂, still correlates 0.79 with NO₂: satellite NO₂ carries genuine emission information. With the CO label, the CO feature is dropped. Because the labels are a ~5 km field (about 11 independent areas), cross-validation blocks must be at least 5 km.

The experiments (Phases 4–6): tree models (XGBoost, Random Forest) with a ridge-regression floor; spatial blocks, cluster hold-out and season hold-out; bootstrap intervals on the differences between experiments; a shuffled-feature negative control (D5). Each experiment is run on both labels:

Experiment	Inputs	Question
A	TROPOMI NO ₂ only	Baseline: the base paper's single-tracer approach on the corridor
B	A + TROPOMI CO (proposed; the proposal said OCO-3, which is too sparse to be a per-cell input, D11)	Does a second tracer of combustion type add skill?
C	B + weather (wind, boundary layer, temperature, sunlight, HCHO)	Does transport and chemistry context help?
D	C + human activity (night lights, roads, NDBI, NDVI, industrial land)	Does ground context separate industry from traffic?

9. Phase 4: a bigger training area and a fair test

A critical review found that the corridor alone is too small for an honest machine-learning test: the labels are a smooth ~5 km field, so neighbouring cells are near-copies. The training area was therefore **enlarged to 2,894 cells** within 30 km of the source (D22), while the corridor stays the evaluation focus. Every corridor value came out identical on both grids, proving the larger pipeline changes nothing that was already there.

	Corridor only	Regional grid (D22)
Cells	268	2,894
How far label errors stay correlated	9.5 km	14.7 km
Cross-validation blocks	10 km (9 blocks)	15 km (23 blocks)
Independent areas (the honest sample size)	~3	~13
NO ₂ -label noise ceiling	0.97	0.99

How big must the test blocks be? (D23) A variogram measures how different two cells are as a function of their distance. The labels never level off (they are smooth at every scale), so they cannot set a block size. But the smooth signal is already handled: every experiment is compared with a model that knows only location. What blocks must stop is neighbouring cells sharing the same label *errors*, and those stop being correlated after about 15 km.

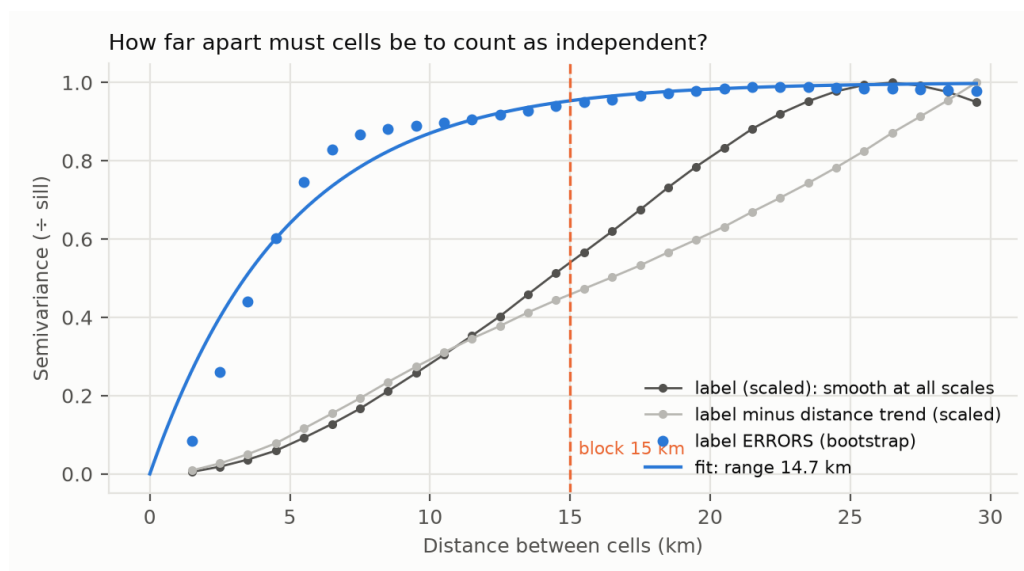


Figure 22. Label errors (blue) decorrelate within ~15 km; the labels themselves (grey) keep rising: smooth at all scales.

The analysis plan (docs/phase5_analysis_plan.md) fixes the experiments, a geography-only null model, a shuffled control, four split schemes, the metrics and the decision rule (a difference counts only if its 95% interval excludes zero), plus seven written predictions, before any model is trained. It is committed to git first, so its timestamp proves the rules came before the results.

10. All design decisions

All 23 decisions are recorded with their evidence in docs/decisions.md. Status: **A** = adopted, **P** = proposed, awaiting the guide's sign-off.

#	Decision	Why	
D1	Estimate emissions per cluster, not per waypoint	Waypoints are closer together than one TROPOMI pixel	A
D2	A layer used to build a label may not also be a model input	Otherwise the model learns the recipe (circularity)	A
D3	OCO-3 as an independent observed target	Only real CO ₂ observation; weight set by G1 (later replaced by D8/D11)	A→D8
D4	Add TROPOMI CO as a source-type tracer	CO:NOx differs by sector	A
D5	Tree models instead of a U-Net; spatial cross-validation	~270 cells × 40 months is too small and autocorrelated for deep learning	A
D6	ERA5 hourly winds matched to each overpass	Finer than MERRA-2	A
D7	October 2019 – May 2024, October–May only	After TROPOMI's pixel change; avoids monsoon cloud	A
D8	OCO used at city scale on its 8 dates, not per cell	Too sparse for a per-cell feature	P
D9	Corridor on the real highway + Talegaon MIDC	Proposal waypoints up to 4.4 km off	A
D10	Plume fit for the total; flux divergence for shares	FD totals depend on lifetime; shares are robust	A
D11	OCO is an upper limit, not a measurement	Pune's CO ₂ plume is 10–20× below noise	P
D12	CO ₂ :NOx ratio constrained by TROPOMI CO (163, 148–180)	Largest error term (25% → ~14%)	P
D13	Outlier filter: each pixel vs its own history, k = 3	Other variants clipped the plume or removed 29% of data	A
D14	Chemistry proxy: HCHO + temperature + sunlight	Reanalysis ozone is ~50 km; TROPOMI O3 is mostly stratospheric	A
D15	1 km UTM grid, cells ≥ 50% inside the corridor (268)	True 1 km squares; loses almost no area	A
D16	Phase 3 labels: seasonal FD label + independent CO label	Avoid circularity; monthly fits too noisy	P
D17	Report an annual-mean equivalent	Like-for-like comparison with annual inventories	A
D18	Monte Carlo uncertainty; method bias handled explicitly	Honest, non-symmetric ranges	A
D19	Fit ≤ 45 km downwind with a sloped background	A second source biased NOx ~20% low	P
D20	Roads: GRIP4 fallback (v1), OpenStreetMap (v2)	OSM servers overloaded; solved with a bulk file	A
D21	CO label is spatial-only	CO seasonal totals vary less than their noise (0.49)	P
D22	Train on a regional grid (2,894 cells); evaluate on the corridor	The corridor alone has ~3 independent areas	A
D23	15 km test blocks, sized by how far label errors spread	Labels are smooth at all scales; errors are what leak	A

11. What went wrong, and how it was caught

Forty problems are logged in the research log and learning guide. The ones that changed the science:

Problem	How it was caught	Fix
Proposal waypoints up to 4.4 km off; Talegaon MIDC outside the corridor	Checked against OpenStreetMap (G4)	Corridor rebuilt on the highway (D9)
Coverage check counted images, not days (439 'overpasses' in a month)	Impossible number	Count distinct days
Plume fit biased +12%	Synthetic test	Causes measured; kept in the budget
ODIAC 2.6× EDGAR	Surprising number → units checked	Tonnes of carbon → ×44/12; real 3.2× disagreement remains
Expected OCO to confirm CO ₂	Predicted plume vs noise	Reported as an upper limit (D11)
Assumed road transport is a high-CO sector	EDGAR's own ratios	Groups defined by the data
CO map peaked over the coast, not Pune	Map inspection	Terrain normalisation + static-pattern removal
Outlier filter removed 29% of real data	Real-data removal rate checked	Per-pixel history filter: 0.09%
Phase 1 declared complete too early	Checked against the proposal's list	Three missing items done
Fit window included a second source	Robustness test changed NO _x by 42%	D19; everything rerun; one conclusion withdrawn
Lockdown plume fit degenerate	R ² 0.62, 8.8 h lifetime	Model-free check instead
Synthetic CO test too easy (random winds)	Re-run with Pune's real winds	Static-pattern removal
Emission profiles matched by name, silently wrong	Printed what was matched	Matched on IPCC codes
Earth Engine outputs blank; ERA5 missing for 7,560 rows	Missing-data audit	Named outputs; sample at 1 km
OSM servers down for days	Progress stuck at 16/35 tiles	Bulk Geofabrik file
A 'share' above 100%	A fraction can't exceed 1	Consistent variance definition
NDBI highest in rural cells	Per-cluster table vs trusted layers	Documented as a caveat

Lessons that shaped the work

- Test every method on fake data with a known answer before trusting it, and use realistic conditions (Pune's real winds).
- Check outputs against physics: 439 'overpasses' in a month, 29% 'outliers' or a 2.6× unit mismatch were all bugs.
- When a method is fixed, rerun everything downstream and re-check every conclusion.
- Report null results honestly: OCO's non-detection is a finding, not a failure.
- When a server is the bottleneck, look for a bulk download of the same data (OpenStreetMap: days of failures vs 2.5 minutes).
- Write everything down: a research log, a decision record and a findings document were kept throughout.

12. Limitations (stated openly)

Limitation	Effect	How it is handled
Midday, October–May observations only	Not a true annual total	Annual mean via published emission profiles ($\pm 10\%$)
Plume-fit bias +12% (synthetic)	Total may be slightly high	In the budget; bias-corrected variant reported
Choice of wind level	0.56–0.67 kg/s across levels	850 hPa justified; sensitivity reported
Corridor zones are only 1–2 TROPOMI pixels wide	Emission smears across zone edges	Shares used; 1 km map is a downscaling
Central Pune (largest source) lies outside the corridor	Transported NO ₂ enters the corridor	Separate Pune-core zone
NO _x /NO ₂ = 1.32 assumed	Scales all NO _x	± 0.1 in the budget (7.6%)
CO-constrained ratio still uses EDGAR's per-sector ratios	Residual ratio error	Assumed 10% term
OCO cannot see the Pune plume	No direct CO ₂ test	Upper limit only
Inventories disagree 3.2×	No reliable 'truth'	Both reported; neither used as truth
OpenStreetMap is a 2026 snapshot	Roads/industry assumed constant 2019–24	Stated assumption
CO, HCHO, weather barely vary between 1 km cells	They can only explain timing	Documented before modelling; spatial CV
NDBI confused by dry bare soil	'Built-up' highest in rural C3	Caveat for Experiment D
Labels are constructed, not observed	Risk of circularity	D2/D16 rules; independent CO label

13. Where the project stands, and what comes next

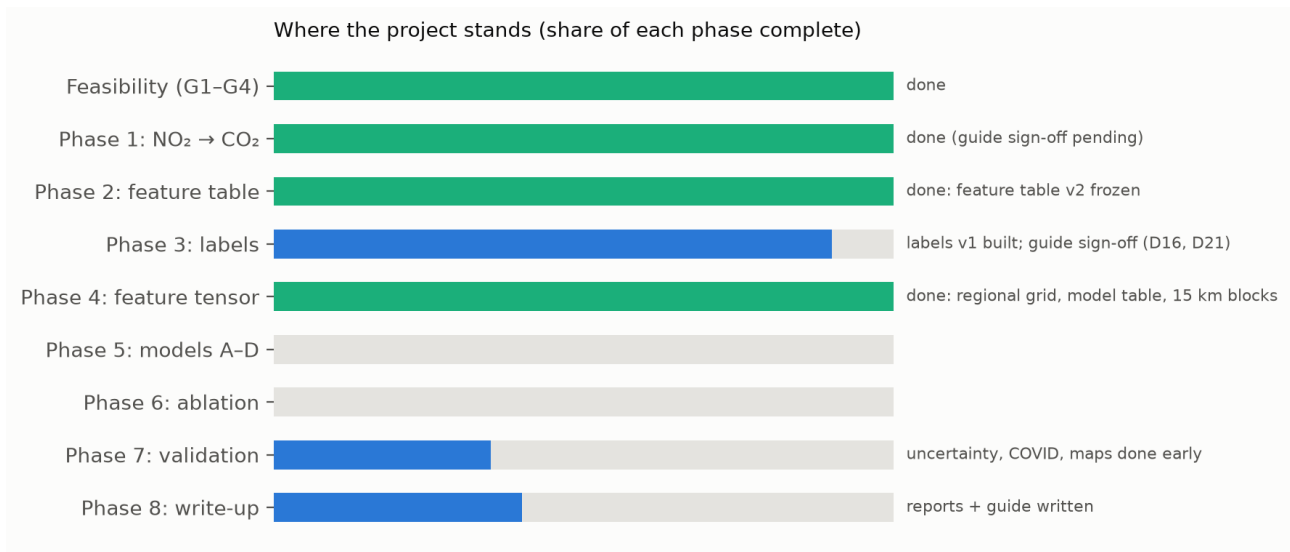


Figure 23. Progress by phase.

Next steps

- **Guide review and sign-off** of the open decisions: OCO's role (D8/D11), the CO-constrained ratio (D12), the fit-window correction (D19), the Phase 3 label design (D16) and the spatial-only CO label (D21).
- **Freeze the analysis plan** (commit it to git) before any model is trained.
- **Phases 5–6, machine learning:** Experiments A–D (NO₂ only → + CO → + weather → + human activity) on both labels, against the geography-only null, on 15 km blocks, the corridor-transfer and PCMC hold-out splits; feature-group ablation; robustness at 10 and 20 km blocks.
- **Phase 7–8:** validation maps, final uncertainty, and the final report.

Questions for the guide: (1) Is the NO₂-based label acceptable as primary, with Experiment A reported as a construction baseline? (2) Should the ablation run on both labels, with the CO label as the NO₂-independent check? (3) Is a spatial-only CO label (D21) acceptable? (4) Is seasonal resolution (1,340 labels) enough? (5) Should Experiment B be redefined around TROPOMI CO, since OCO cannot be a per-cell feature?

Full detail: docs/phase1_report.md ... docs/phase4_report.md, docs/phase5_analysis_plan.md, docs/findings.md, docs/decisions.md, docs/phase3_design.md, and the learning guide in docs/guide/.

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- Data: Copernicus Sentinel-5P TROPOMI; ECMWF ERA5; NASA OCO-2/OCO-3 Lite v11 (GES DISC); NOAA VIIRS DNB; Copernicus Sentinel-2; SRTM; OpenStreetMap contributors (Geofabrik extract, ODbL).

Appendix: how to reproduce everything

All settings live in configs/study.yaml. From the project folder, with the virtual environment active, run in order:

- **Feasibility:** g4_verify_waypoints, g4_map, g2_tropomi_coverage, g3_no2_signal, g1_oco_soundings (python -m ecotrack.feasibility.<name>), then acquire.era5_overpass.
- **Phase 1:** acquire.tropomi_cube; inversion.run_city; inversion.run_divergence; acquire.edgar, acquire.odiad; inversion.run_co2; inversion.run_oco_check; acquire.tropomi_cube --product co; acquire.dem; inversion.run_co_ratio; inversion.de_sensitivity; inversion.run_seasonal; inversion.run_co_divergence; inversion.temporal_adjust; inversion.mc_budget.
- **Phase 2:** grid; acquire.grid_layers_gee; acquire.roads_grip; acquire.osm_pbf (after downloading the Geofabrik file); features.
- **Phase 3–4:** labels; tensor. **Regional grid (D22):** set ECOTRACK_GRID=region and rerun grid, grid_layers_gee, roads_grip, osm_pbf, features, labels and tensor.
- **Tests:** python -m pytest (17 tests). **This report:** python -m ecotrack.progress_report.